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EP 1108 (Modern Physics) Final exam

You have 1 hour to submit your answers. Each question has one or more correct answers. You have to select all the correct ones and none of the wrong ones to get credit. Please save screenshots of your answers before submitting.

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Name *

Pushkal Mishra

Roll number *

EE20BTECH11042

(3 points) One gm of radium (mass number 226) has an activity equal to 1 curie. What is the approximate half-life of Radium in years? (Avogadro number = 6 x 10^[23])

- [x] 1600
- [ ] 160
- [ ] 1.6
- [ ]

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(3 points) One gm of radium (mass number 226) has an activity equal to 1 curie. What is the approximate half-life of Radium in years? (Avogadro number = 6 x 10^[23])

- [x] 1600
- [ ] 160
- [ ] 1.6
- [ ] 0.0016

(3 points) Two protons (with rest mass m) moving towards each other with equal and opposite velocity (v), collide and produce two protons and another particle of rest mass M. What is energy of each proton (before the collision) if all the reactant particles are produced at rest in the C.O.M frame after the collision? (Assume c=1)

- [ ] 2m+M
- [ ] m+M/2
- [ ] M
- [x] m+2M+(M^2/2m)

(1 point) Two nuclei (A and B) mass numbers 2 and 128 respectively. The ratio of their radii (radius of nucleus A over nuclei B) is equal to

- [x] 1:4
- [ ] 1:8

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(1 point) Two nuclei (A and B) mass numbers 2 and 128 respectively. The ratio of their radii (radius of nucleus A over nuclei B) is equal to

- [x] 1:4
- [ ] 1:8
- [ ] 1:64
- [ ] None of the above

(1 point) Two nuclei X and Y have mass numbers equal to 23 and 25 and atomic number equal to 12 and 14, respectively. These nuclei can be considered as

- [ ] Isotopes
- [ ] Isobars
- [x] Isotones
- [ ] None of the above

(1 point) A particle of rest mass m has (relativistic) K.E. equal to 3mc^2. So total momentum of the particle is equal to?

- [ ] sqrt(6) mc
- [ ] 2 mc
- [x] sqrt(15)mc
- [ ] sqrt(3)mc

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(1 point) A radioactive element has a half-life of 20 years. What fraction of nuclei (from this element) would have decayed after 40 years?

- [ ] 1/4
- [x] 3/4
- [ ] 1/2
- [ ] Cannot be determined

(3 points) An elementary particle has a rest lifetime of 10^{-5} secs. What is the total distance travelled (as seen by an observer at rest if was created with a speed of 0.99c (where c=3 x 10^8 m/sec)

- [x] 21300 m
- [ ] 3000 m
- [ ] 30,000m
- [ ] 30 m

(1 point) Photons obey which of the following statistics/laws?

- [ ] Maxwell Botzmann statistics
- [ ] Fermi Dirac statistics
- [ ] Bose Einstein statistics
- [x] Planck's law

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Question

(3 points) In a canonical ensemble, the energy of the microstates is allowed to fluctuate keeping the temperature constant. We have shown in class that the average energy is $\langle E\rangle = -(\partial(\ln Q)/\partial\beta)|_{N,V}$. Now the fluctuation in energy is defined as $\Delta E = E - \langle E\rangle$. Which of the expressions below give the correct average energy fluctuation square, $\langle(\Delta E)^2\rangle$?

a) $\langle(\Delta E)^2\rangle = \langle E^2\rangle - 2\langle E\rangle^2$

b) $\langle(\Delta E)^2\rangle = \langle E^2\rangle - \langle E\rangle^2$

c) $\langle(\Delta E)^2\rangle = \frac{\partial^2 \ln Q}{\partial\beta^2}|_{N,V} - (\frac{\partial \ln Q}{\partial\beta})^2|_{N,V}$

d) $\langle(\Delta E)^2\rangle = \frac{\partial^2 \ln Q}{\partial\beta^2}|_{N,V}$

- [ ] a)
- [x] b)
- [x] c)
- [ ] d)

(3 points) In a laser cavity there are 10^10 stimulated emissions occurring per second per cm^3. Assume that in a steady state operation all of the energy of these stimulated emissions goes into laser light. The energy level separation of the lasing levels is 20eV and the length of the optical cavity is 20cm. What is the intensity of the laser beam?

- [x] 0.64 microWatts/cm^2
- [ ] 0.23 microWatts/cm^2
- [ ] 1.5 microWatts/cm^2
- [ ] 92 nanoWatts/cm^2

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- [x] 0.64 microWatts/cm^2
- [ ] 0.23 microWatts/cm^2
- [ ] 1.5 microWatts/cm^2
- [ ] 92 nanoWatts/cm^2

Question

(3 points) Consider a grand canonical ensemble of a quantum system with 2 quantum states. These two states have energies of 1.1×10^{-3}eV and 1.3×10^{-3}eV. The system is populated with bosons and their chemical potential is 10^{-3}eV. At a temperature of 100K, find the average number of particles and energy of this ensemble.

a) $\langle N \rangle = 86, \langle E \rangle = 0.12eV$
b) $\langle N \rangle = 114, \langle E \rangle = 0.13eV$
c) $\langle N \rangle = 57, \langle E \rangle = 1.2 \times 10^{-3}eV$
d) $\langle N \rangle = 28, \langle E \rangle = 1.2 \times 10^{-3}eV$

- [ ] a)
- [x] b)
- [ ] c)
- [ ] d)

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Question

(1 point) Consider a straight chain of N atoms with spins, each spin pointing either up or down. If two neighboring spins are aligned (i.e., pointing in same direction), the energy of the pair is 0 and if they are misaligned, energy is ϵ. The total energy will be $n\epsilon$, with n the number of neighboring pairs that are misaligned. What is the number of micro-states as a function of n, $\Omega(n)$?

a) ${}^{(N-1)}C_n$
b) $n! \times {}^{N}C_n$
c) $2 \times {}^{(N-1)}C_n$
d) ${}^{N}C_n$

- [x] a)
- [ ] b)
- [ ] c)
- [ ] d)

Question

(1 point) Consider a system which has a free energy given by

$$G = \begin{cases} A(T - T_0)^2 \text{ for } T < T_0 \\ B(T - T_0)^3 \text{ for } T > T_0 \end{cases}$$

where A and B are non-zero constants and T is temperature. Which of the following statements are true?

- [ ] There is a zeroth order phase transition at T=T0
- [ ] There is a first order phase transition at T=T0
- [x] There is a second order phase transition at T=T0
- [ ] There is no phase transition

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(1 point) As a ferromagnetic substance is heated, which of the following happens?

- [x] It transitions to a paramagnetic substance
- [ ] It transitions to a diamagnetic substance
- [ ] Its entropy changes discontinuously at the Curie temperature
- [x] Its entropy changes continuously at the Curie temperature

(1 point) What is the average kinetic energy of an ideal gas atom at 300K?

- [x] 0.039eV
- [ ] 0.013eV
- [ ] 0.026eV
- [ ] 7.3X10^(-20)J

(1 point) In a Stern-Gerlach experiment with a spin 3/2 particle the following will be observed

- [x] Four states
- [x] +3/2, +1/2, 0, -1/2, -3/2
- [x] No z-component of spin is zero
- [ ] Five states

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(1 point) The 1-D potential is V(x)= x^2 cos(x), what is the expectation value of x?

- [ ] 1
- [ ] x
- [ ] x^2
- [x] 0

(1 point) A quantum particle with energy E at high temperature will follow which distribution?

- [x] FD
- [ ] BE
- [ ] FD and BE
- [x] MB

(1 point) Consider a 2-D infinite well potential. What is the degeneracy of the first excited state?

- [ ] 1
- [ ] 2
- [x] 3
- [ ] 0

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follow relations are true.

- [x] Ef=n^{2/3}
- [x] Electron distribution function=1 for E < Ef
- [x] Electron distribution function =1 for E > Ef
- [ ] nf= Ef

(3 points) Consider a particle in a 1-D infinite potential well between x=0, L. Calculate the probability of the particle to be within x=0, L/a and the followings are true. Usual notations are used.

- [ ] 1/a + Sin(2*n*pi*/a)/(2*n*pi)
- [ ] 1/a
- [x] 1/a - Sin(2*n*pi*/a)/(2*n*pi)
- [x] 1/a for large n

(3 points) Consider 1 g of marble inside 1 cm long tube. Consider this is as a 1-D infinite potential well and calculate the quantum number n for the energy E=1 mJ and the exception energy to promote it to the next level. The symbols are usual.

- [x] 4.27 *10^{28}, 4.69*10^{-32} J
- [ ] 4.27 *10^{-28}, 1m J
- [ ] 4.27 *10^{20}, 4.69*10^{-32} J
- [ ] 1000.0, 4.69*10^{-32} J